

Motivation: If instead of the categorized mix data, we were presented with the 150 observations in an unclassified manner, then the aim might have been to group the data into homogeneous classes or clusters.

Single-Sample Case: Let us assume that $x_1, \dots, x_n$ are independent. Further each may arise from any one of g possible sub-populations with p.d.f. $f(x, \theta_k)$; $k=1, \dots, g$, where we assume g is known.

Let $\mathcal{V} = (v_1, \dots, v_n)^T$ be a set of identifying labels so that $v_i = k \Rightarrow x_i$ comes from k th sub-population $i=1, \dots, n$; $k=1, \dots, g$. Suppose that $C_k =$ the set of x_i assigned to the k th group by $\mathcal{V}$, $k=1, \dots, g$ The likelihood function is

$$L(\mathcal{X}; \theta_1, \dots, \theta_g) = \prod_{x \in C_1} f(x; \theta_1) \dots \prod_{x \in C_g} f(x; \theta_g) \longrightarrow \textcircled{1}$$

The normal case: Let us assume that $f(\underline{x}, \theta_k)$ denotes the pdf of $N_p(\underline{\mu}_k, \Sigma_k)$, $k=1, \dots, g$. Then the likelihood function is

$$l(\underline{\mathcal{L}}; \underline{\theta}) = \text{const} - \frac{1}{2} \sum_{k=1}^g \sum_{\underline{x}_j \in C_k} (\underline{x}_j - \underline{\mu}_k)^T \Sigma_k^{-1} (\underline{x}_j - \underline{\mu}_k) - \frac{1}{2} \sum_{k=1}^g n_k \log |\Sigma_k|,$$

where there are n_k observations in C_k .

Hence for given $\underline{\mathcal{L}}$, the MLE are

$$\hat{\underline{\mu}}_k(\underline{\mathcal{L}}) = \bar{\underline{x}}_k, \quad \hat{\Sigma}_k(\underline{\mathcal{L}}) = S_k. \quad \text{--- (2)}$$

where $\bar{\underline{x}}_k$ is the mean and S_k is the covariance matrix of the n_k observations in C_k . On substituting

(2) in (1), we obtain

$$l(\underline{\mathcal{L}}; \hat{\underline{\theta}}(\underline{\mathcal{L}})) = \text{const} - \frac{1}{2} \sum_{k=1}^g n_k \log |S_k|.$$

Hence the MLE of $\underline{\mathcal{L}}$ is the grouping that minimizes

$$\prod_{k=1}^g |S_k|^{n_k}. \quad \text{--- (3)}$$

If we assume that the groups have identical covariance matrices, $\Sigma_1 = \dots = \Sigma_k = \Sigma$, the same method leads to the grouping that minimizes

$$|W|$$

→ (4)

where
$$W = \sum_{k=1}^g \sum_{x_j \in C_k} (x_j - \bar{x}_k)(x_j - \bar{x}_k)^T,$$

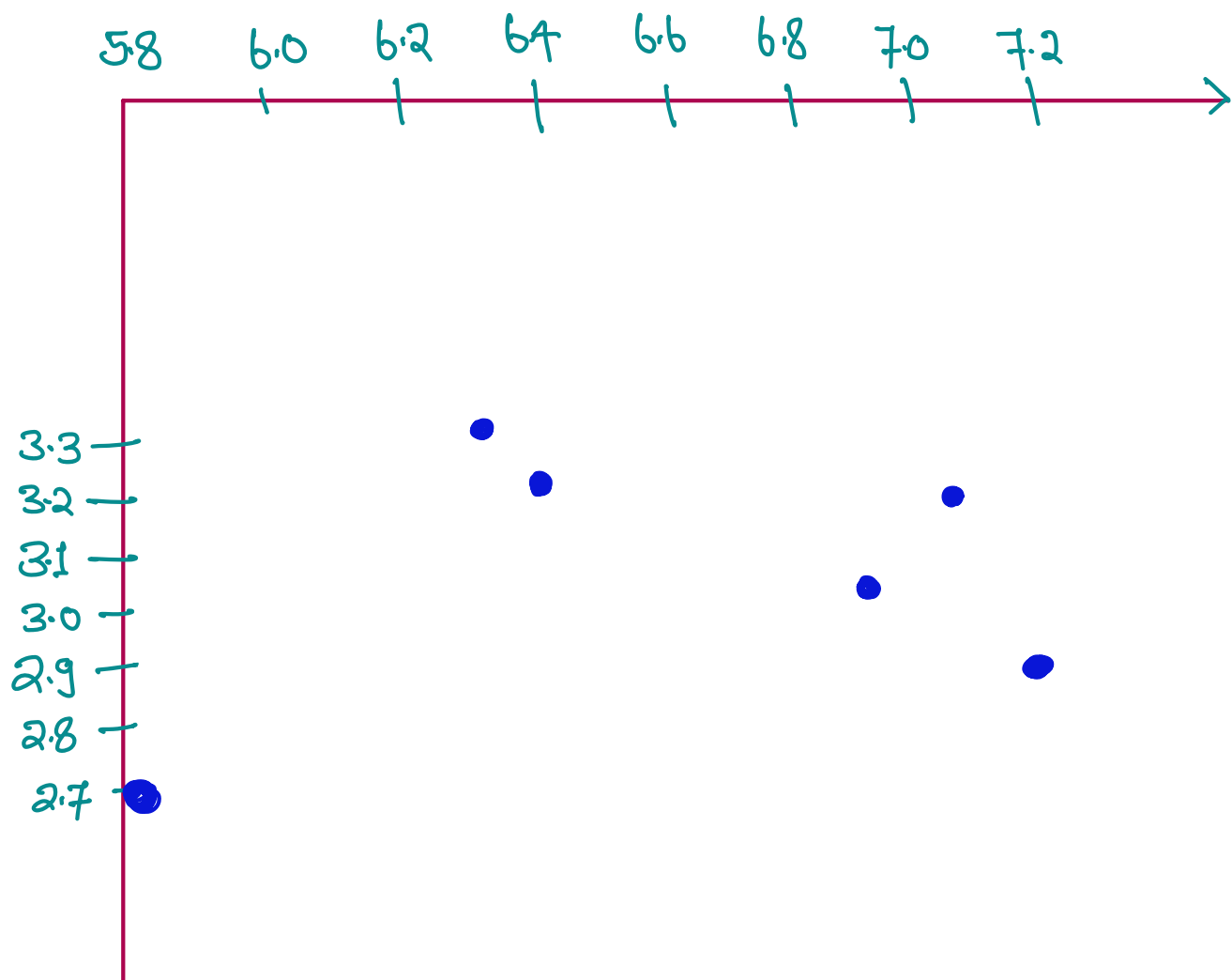
is the pooled within-groups sums of squares and products (SSP) matrix.

Example:

A subsample of iris data

	1	2	3	4	5	6
Sepal length	7.0	6.4	6.9	6.3	5.8	7.1
Sepal width	3.2	3.2	3.1	3.3	2.7	3.0

Scatter plot:



G_1	G_2	$ S_1 $	$ S_2 $	$10^3 S_1 S_2 $	$ S_1 + S_2 $
(123)	(456)	0.001 200	0.132 300	0.1588	0.180
(124)	(256)	0.001 200	'	'	'
(125)	(346)	0.030 000	'	'	'
(126)	(345)	0.004 800	'	'	'
'	'	'	'	'	'
(136)	(245)	0.000 300	0.004 033	0.0012	0.017
'	'	'	'	'	'
(156)	(234)	0.028 033	0.000 533	0.0150	0.114

↑

Unequal

$\Sigma_1 \neq \Sigma_2$

↑

Equal

$\Sigma_1 = \Sigma_2$

In both cases, ML method leads to the clusters (1,3,6) and (2,4,5).

Multi-Sample Case: For the multi-sample problem, we shall consider the normal samples with equal covariance matrix. Each sample can be summarized by its mean $\bar{x}_i$ and SSP matrix W_i . Then $\bar{x}_i \sim N_p(\mu_i, n_i^{-1}\Sigma)$, independently for $i=1, \dots, m$, and a consistent estimate of Σ is given by

$$\hat{\Sigma} = W / (n-m) \quad \text{--- (5')}$$

It would be good if we can get a better understanding of multi-sample data.

Example: White toothed shrews occur in the Channel and Scilly Islands of the British Isles and French mainland.

From $p=10$ measurements on each of $n=399$ skulls obtained from the $K=10$ localities it is believed that there may be two species of shrews.

The localities were demographically close, but it is assumed that only one sub-species was present in any one place.

Thus the problem here is to group "samples" rather than "objects" as in the single-sample case.

Multi-Sample Problem: Let x_{ij} , $i=1, \dots, n_j$ be the observations in the j th sample, $j=1, \dots, m$. The aim of cluster analysis is to group the m samples to g homogeneous classes.

Let's go back to ⑤. W is defined by

$$W = \sum_{j=1}^m \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)(x_{ij} - \bar{x}_j)^T \quad \leftarrow \text{Within Sample SSP matrix}$$

where $\bar{x}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} x_{ij}$.

Let $\bar{x} = \frac{1}{n} \sum_{j=1}^m \sum_{i=1}^{n_j} x_{ij}$, $n = \sum_{j=1}^m n_j$

Suppose these m samples have come from only g population ($g \leq m$) i.e;

$$\mu_i \in (\mu_1, \dots, \mu_g) \quad g \leq m.$$

Thus there are only g distinct means out of $\mu_1, \dots, \mu_m$.

Let $\mathcal{L}$ be the set of identifying labels, and let C_k be the set of $\bar{x}_j$ assigned to the k th group by $\mathcal{L}$.

$k=1, \dots, g$. Now maximizing the likelihood implies minimizing (Σ is assumed to be known)

$$\sum_{k=1}^g \sum_{\underline{x}_j \in C_k} (\underline{\bar{x}}_j - \underline{\mu}_k)^T \Sigma^{-1} (\underline{\bar{x}}_j - \underline{\mu}_k) \quad \text{--- (6)}$$

Let
$$N_k = \sum_{\underline{x}_j \in C_k} n_j, \quad \underline{\bar{\bar{x}}}_k = \frac{\sum_{\underline{x}_j \in C_k} n_j \underline{\bar{x}}_j}{N_k}$$

so that $\underline{\bar{\bar{x}}}_k$ is the weighted mean vector of the means in the k th group. From (6) $\hat{\underline{\mu}}_k = \underline{\bar{\bar{x}}}_k$, so the MLE of $\underline{\mu}$ is the grouping that minimizes

$$W_g^2 = \sum_{k=1}^g \sum_{\underline{x}_j \in C_k} n_j (\underline{\bar{x}}_j - \underline{\bar{\bar{x}}}_k)^T \Sigma^{-1} (\underline{\bar{x}}_j - \underline{\bar{\bar{x}}}_k). \quad \text{--- (7)}$$

Let

$$D_{ij}^2 = (\underline{\bar{x}}_i - \underline{\bar{x}}_j)^T \Sigma^{-1} (\underline{\bar{x}}_i - \underline{\bar{x}}_j)$$

is the Mahalanobis distance between $\underline{\bar{x}}_i$ and $\underline{\bar{x}}_j$. It can be shown that

$$W_g^2 = \sum_{k=1}^g N_k^{-1} \sum_{\underline{\bar{x}}_i, \underline{\bar{x}}_j \in C_k} n_i n_j D_{ij}^2 \quad \text{--- (8)}$$

Example: Shown data: $K=10$.

$$n_1 = 144, n_2 = 16, n_3 = 12, n_4 = 7, n_5 = 90$$

$$n_6 = 25, n_7 = 6, n_8 = 26, n_9 = 53, n_{10} = 20.$$

We assume a priori that we are looking for two clusters C_1 and C_2 with 3 and 7 elements, respectively.

We wish to minimize w_j^2 over 120 partitions.

Partition C_1	w_j^2
(2, 9, 10)	643
(7, 2, 9)	1997
⋮	
(1, 8, 10)	5546

For unknown Σ , you can use a consistent estimate of Σ say $\hat{\Sigma} = W/n-m$

Heuristical Methods:

Nearest neighbour single linkage cluster analysis

Let $D = (d_{ij})$ is the distance matrix between the points $x_1, \dots, x_n$. This method groups the points which are nearest to one another.

Example: Consider the given data. There are $k=10$ clusters of size 1 (for each location sample mean is considered as a data point).

Calculate

d_{ij} for all $i \neq j \in \{1, \dots, 10\}$.

Arrange them in increasing order.

$d_{12}, d_{13}, \dots, d_{9,10}$. Find the minimum of d_{ij} say d_{35} . Then club these two clusters.

$$d_{35} = 1.35.$$

(1) (2) (3, 5) (4) (6) ... (10)

Next look for second minimum of d_{ij} which is d_{15}

$(1, 3, 5), (2), (4), (6) \dots (10)$.

next minimum dist

$(1, 2, 3, 5) (4) (6) \dots (10)$

d_{25}
:
i

Steps:

- a) Let $C_1, \dots, C_n$ be starting clusters each containing one point, namely $x_i \in C_i$.
- b) Without any loss of generality, let $d_{x_1 x_2} = \min d_{ij}$ so that x_{i_1} and x_{j_1} are nearest. Then these two points are grouped into a cluster, so we have $n-1$ clusters, where $C_{i_1} + C_{j_1}$ is a new cluster.
- c) Let $d_{x_2 x_3}$ be the next smallest distance. If neither x_1 nor x_2 equals x_2 or x_3 , the new $n-2$ clusters are $C_{i_1} + C_{j_1}, C_{i_2} + C_{j_2}$ plus the remaining old clusters. If $x_2 = x_1$ and $j_1 \neq j_2$ the new $n-2$ clusters are $C_{i_1} + C_{j_1} + C_{j_2}$, plus the remaining old clusters.

d) The process continues as in c) through all $\frac{1}{2}n(n-1)$ distances. At the i -th stage let $d_{i,i'}$ denote the smallest distance. Then the cluster containing x_i is joined with the cluster containing $x_{i'}$. Note that if x_i and $x_{i'}$ are already in the same cluster, then no new groups are formed in this stage.

e) The clustering process can be halted before all the clusters have been joined into one group by stopping when the inter-cluster distances are all greater than d_0 , where d_0 is an arbitrary threshold level.

Distances and Similarities:

Definition: Let P and Q two points where there may represent measurements x and y on two objects. A real valued function $d(P, Q)$ is a distance function if it has the following properties:

I) Symmetry, $d(P, Q) = d(Q, P)$.

II) Non-negativity $d(P, Q) \geq 0$.

III) Identification max $\mathbb{R}$ $d(P, P) = 0$.

For many distance functions the following properties also hold:

IV) definiteness, $d(P, Q) = 0$ if and only if $P = Q$;

V) triangle inequality, $d(P, Q) \leq d(P, R) + d(R, Q)$.

If (I) - (V) hold d is called a metric.

Quantitative data:

Euclidean Distance: Let $X_{n \times p}$ matrix with rows $\underline{x}_1^T, \dots, \underline{x}_n^T$. Then the Euclidean distance between the points $\underline{x}_i$ and $\underline{x}_j$ is d_{ij} where

$$d_{ij}^2 = \sum_{k=1}^p (x_{ik} - x_{jk})^2 = \|\underline{x}_i - \underline{x}_j\|^2$$

This distance function satisfies properties (I) - (V). It also satisfies the following properties:

Mahalanobis Distance:

$$D_{ij}^2 = (\underline{x}_i - \underline{x}_j)^T \Sigma^{-1} (\underline{x}_i - \underline{x}_j)$$

Complete LinBage:

$$d_{35} = 1.35$$

$$d_{15} = 1.40$$

$$d_{12} = 1.61$$

$$d_{25} = 1.70$$

$$d_{14} = 1.969$$

$$d_{13} = 1.972$$

$$d_{23} = 2.02$$

$$d_{45} = 2.21$$

$$d_{16} = 2.45$$

$$d_{24} = 2.51$$

$$d_{9,10} = 2.56$$

$$d_{8,9} = 2.64$$

$$d_{17} = 2.84$$

$$d_{34} = 2.88$$

$$d_{47} = 2.89$$

$$d_{67} = 3.00$$

⋮

$$d_{49} = 7.08$$

